KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

NAME:

TEACHER:

MATHEMATICS

Year 7

Practice Exam 3

2026

Time allowed: 50 mins
5 mins reading time

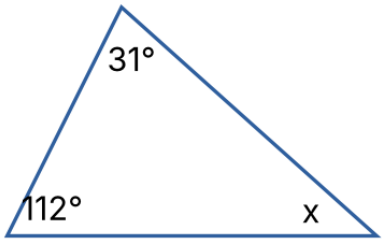
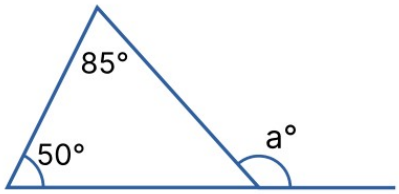
Instructions	**Sections and Marks**
1. Attempt all questions 2. Show working where needed 3. A calculator is **NOT** permitted	Part A (Multiple Choice) – 15 Marks Part B (Calculations) – 30 Marks Part C (Data and Probability) – 18 Marks **Total – 63 Marks**

Part A

1.	Which word describes the difference between the highest and lowest values in a set of data? a) Mean b) Mode c) Range d) Median	
2.	The median of 8, 3, 5, 9, 1 is: a) 3 b) 5 c) 8 d) 9	
3.	Which calculation follows the order of operations correctly? a) $2 + 3 \times 4 = 20$ b) $2 + 3 \times 4 = 14$ c) $(2 + 3) \times 4 = 14$ d) $2 \times (3 + 4) = 10$	
4.	Which shows 0.5, 0.05, 5, 0.55 in descending order? a) 5, 0.55, 0.5, 0.05 b) 0.05, 0.5, 0.55, 5 c) 5, 0.5, 0.55, 0.05 d) 0.55, 5, 0.5, 0.05	
5.	In the number 9.207, the digit 7 is in which place? a) Tenths b) Hundredths c) Thousandths d) Ten thousandths	
6.	What is the value of 3^3? a) 9 b) 27 c) 6 d) 33	
7.	Which of the following contains a variable? a) $12 + 5$ b) 3×7 c) $y + 4$ d) $20 \div 2$	
8.	Which set of angles could be the three angles of a triangle? a) 50°, 60°, 80° b) 90°, 45°, 45° c) 100°, 60°, 30° d) 70°, 70°, 70°	
9.	Which statement about an isosceles triangle is correct? a) It has no equal sides b) It has two equal sides and two equal angles c) All of its angles are 90° d) It has four sides	
10.	A square has sides of 6 cm. What is its perimeter? a) 12 cm b) 24 cm c) 36 cm d) 6 cm	

11.	The temperature is −3 °C and then rises by 5 degrees. The new temperature is: a) −8 °C b) 8 °C c) 2 °C d) −2 °C	
12.	A standard die is rolled. How many outcomes in the sample space are even numbers? a) 2 b) 3 c) 4 d) 6	
13.	Which of these could NOT be a value of P(E)? a) 0 b) $\frac{1}{2}$ c) 1 d) $\frac{3}{2}$	
14.	An angle of exactly 90° is called: a) Acute b) Right c) Obtuse d) Straight	
15.	Angles that form a ‘Z’ shape between parallel lines are called: a) Corresponding b) Co-interior c) Alternate d) Vertically opposite	

Part B

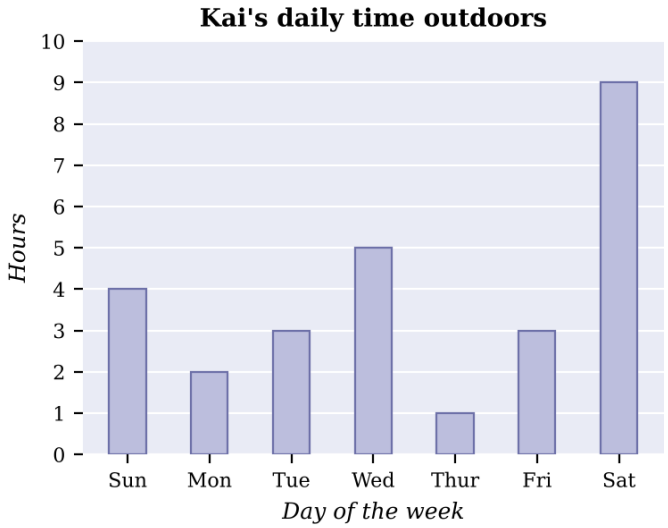
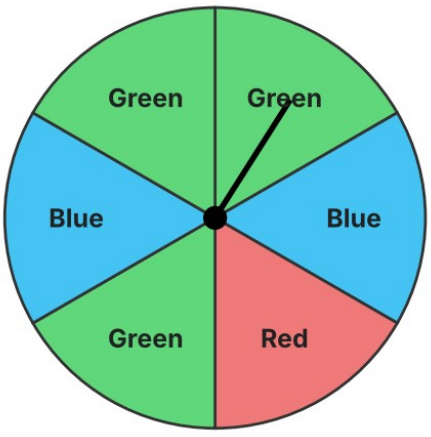
1.	Calculate the missing angle and provide working out.  	2 marks
2.	Calculate the size of the exterior angle marked a° and provide working out.  	2 marks
3.	State True or False for each of the following: a) $\frac{5}{6} = \frac{15}{18}$ b) $\frac{2}{9} = \frac{8}{27}$	2 marks
4.	Calculate the following: a) $0.6 \times 3 =$ b) $\frac{1}{4} \times \frac{3}{5} =$	2 marks
5.	Calculate the following: a) $-15 + 9 =$ b) $-8 - 2 =$ c) $3 \times (-1 + 6) =$	3 marks

6.	Convert into a mixed number: a) $\frac{13}{5}$ Convert into an improper fraction: b) $3\frac{1}{6}$	2 marks
7.	Write an algebraic expression for each statement. Use x to represent the number: i) 7 less than the number ii) Twice the number, plus 1	2 marks
8.	If $g = 6$, $h = 2$ and $j = 5$ Solve the following by substitution: i) $g + h + j =$ ii) $3h + j =$	2 marks
9.	Simplify the following: i) $9b + 4b =$ ii) $2mn + 6mn + mn =$ iii) $7r + 10 - 5r =$ iv) $6 \times 2z =$	4 marks
10.	Calculate the following (use a mental or written strategy): i) $24 + 17 + 6 + 3 =$ ii) $9 \times 600 =$ iii) $31 \times 4 =$	3 marks

11.	Solve the following equations: i) $a + 12 = 20$ ii) $b - 5 = 9$ iii) $4c = 36$ iv) $\frac{e}{6} = 2$ v) $5y + 3 = 28$ vi) $\frac{q - 1}{4} = 3$	6 marks

Part C

12.	The list below shows how many times each student in 7B bought lunch from the canteen in one term: 4 5 3 6 5 4 7 5 2 4 5 6 3 5 14 4 6 5 3 4 i) Construct a dot plot from the above set of data ii) How many students are in the data set? iii) What is the mode? iv) What is the range? v) How many students bought lunch fewer than 4 times? vi) Is there an outlier? If so, what is it?	7 marks
13.	Observe the following set of data: 5, 9, 3, 7, 11, 4, 8, 6, 10 Calculate the following: i) The mean ii) The median If 17 was added to the data set, what would be: iii) The new median? iv) The new mean?	4 marks

14.	Observe the following graph and answer the questions below.  i) How many hours did Kai spend outdoors on Wednesday? ii) On which day did Kai spend the least time outdoors? iii) How many hours did Kai spend outdoors over the whole week shown? iv) What fraction of Saturday did Kai spend outdoors? Give your answer in simplest form.	4 marks
15.	 Observe the spinner above. Each section is the same size. a) List the sample space: b) What is the probability of spinning blue? c) What is the complement of spinning green, and what is its probability?	3 marks